INCH-POUND
MIL-DTL-55302/99D
w/AMENDMENT 1
3 June 2010
SUPERSEDING
MIL-DTL-55302/99D
17 November 2004

DETAIL SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES: RECEPTACLE, STRAIGHT THROUGH, HERMAPHRODITIC CONTACT, WRAPPOST REMOVABLE, FOR PRINTED WIRING BOARDS (.100 INCH SPACING)

Inactive for new design after 24 October 1991.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-55302.

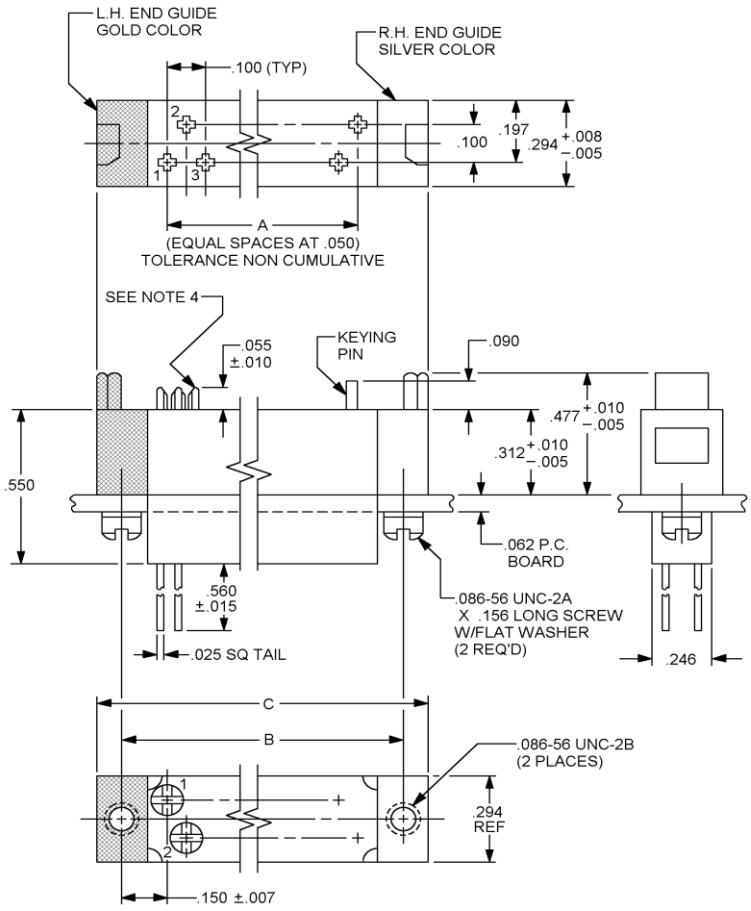


FIGURE 1. Connector, receptacle (.100 inch spacing).

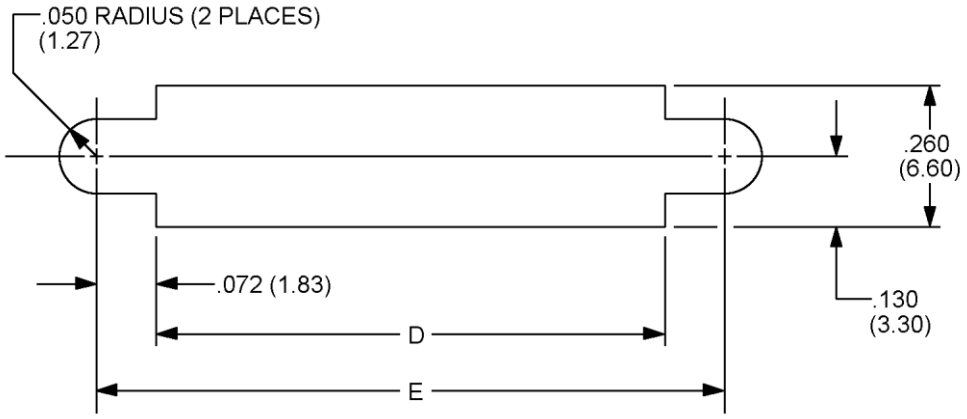
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Inches	mm	Inches	mm
.005	0.13	.130	3.30
.007	0.18	.150	3.81
.008	0.20	.156	3.96
.010	0.25	.197	5.00
.015	0.38	.246	6.25
.025	0.64	.260	6.60
.050	1.27	.273	6.93
.055	1.40	.294	7.47
.062	1.57	.312	7.92
.072	1.83	.477	12.12
.086	2.18	.550	13.97
.100	2.54	.560	14.22

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ± .010 (0.25 mm).
4. With connectors fully engaged the mating contacts shall have a minimum overlap distance of .062(1.57mm). As a minimum the contacts shall exhibit a wiping action through this distance.

FIGURE 1. Connector, receptacle (.100 inch spacing) – Continued.

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NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parentheses.

FIGURE 2. Typical cutout for receptacle in nonconductive chassis.

TABLE I. Dash number and dimensions.

Dash number 1/	Number of contacts	Dimensions 2/				
		A	B ±.010	C Ref	D	E
01	18	.850 (21.59)	1.150 (29.21)	1.300 (33.02)	1.006 (25.55)	1.150 (29.21)
02	30	1.450 (36.83)	1.750 (44.45)	1.900 (48.26)	1.606 (40.79)	1.750 (44.45)
03	36	1.750 (44.45)	2.050 (52.07)	2.200 (55.88)	1.906 (48.41)	2.050 (52.07)
04	42	2.050 (52.07)	2.350 (59.69)	2.500 (63.50)	2.206 (56.03)	2.350 (59.69)
05	54	2.650 (67.31)	2.950 (74.93)	3.100 (78.74)	2.806 (71.27)	2.950 (74.93)
06	72	3.550 (90.17)	3.850 (97.79)	4.000 (101.60)	3.706 (94.13)	3.850 (97.79)

1/ See requirements for complete Part or Identifying Number (PIN) when keying pin is required.
2/ Dimensions are in inches. Metrics are in parentheses.

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REQUIREMENTS

Dimensions and configurations: See figures 1 and 2, and table I.

Material and finish:

Insulator: ASTM D5948, type SDG-F.

End guides: Left hand, sintered brass 90/10, clear chromate in accordance with MIL-DTL-5541, class 1A.

Mounting screw: Brass, cadmium plated, gold chromate finish.

Flat washer: Stainless steel.

Contact identification: Contact locations are identified in numerical sequence (see figure 1).

Mating and unmating: The maximum insertion force, in pounds, shall not exceed a value equal to .5 times the number of contacts, and the withdrawal force, in pounds, shall be a minimum of .11 times the number of contacts and shall not exceed the measured insertion force.

Contact retention: 6.0 pounds minimum after five insertions and withdrawals from the same connector position.

Contact resistance: The contact resistance shall not exceed 20 milliohms.

Dry circuit: Connectors shall be tested in accordance with test procedure EIA-364-23. The contact resistance shall not exceed 20 milliohms.

Dielectric withstanding voltage:

Sea level: 1,000 volts rms, 60 Hz.
High altitude: 300 volts rms, 60Hz.

Current rating, maximum: 5 amperes.

Keying: When required, one of the following codes shall be added to the number:

P – Specifying location by contact number where keying pin shall be inserted; for example, M55302/99-01P17. (Keying pin inserted in place of contact number 17.)

H - Specifying location by contact number where contact shall be omitted for mating; for example, M55302/99-01H17. (Keying pin hole has contact number 17 omitted.)

Contacts: Shall conform to MIL-DTL-55302/98.

Contact removal tool: Shall conform to MIL-I-81969/4.

Contact insertion tool: Shall conform to MIL-I-81969/5.

Mating plug: Shall conform to MIL-DTL-55302/53 and MIL-DTL-55302/54.

Contact wrappost tool: Shall conform to MIL-DTL-55302/98.

PIN: M55302/99-(dash number from table I).

Group submission: MIL-DTL-55302/90, MIL-DTL-55302/92, and MIL-DTL-55302/94.

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Amendment notations. The margins of this specification are marked with vertical lines to indicate where modifications from this amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-DTL-55302, this document references the following:

MIL-DTL-55302/53
MIL-DTL-55302/54
MIL-DTL-55302/90
MIL-DTL-55302/92
MIL-DTL-55302/94
MIL-DTL-55302/98
MIL-DTL-5541
MIL-I-81969/4
MIL-I-81969/5
ASTM D5948
EIA-364-23

CONCLUDING MATERIAL

Custodians:
Army – CR
Air Force - 99
DLA – CC

Preparing activity:
DLA - CC

(Project 5935-2010-120)

Review activities:
Army – AR, AT, AV, CR4, MI

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil.

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